INVERSE LAPLACE TRANSFORM

Definition: If $L\{F(t)\} = f(p)$ then, $F(t)$ is called inverse Laplace Transform of $f(p)$ and is denoted by $L^{-1}\{f(p)\} = F(t)$

e.g. $L\{e^{5t}\} = \frac{1}{p-5} \Rightarrow L^{-1}\left\{\frac{1}{p-5}\right\} = e^{5t}$

Inverse Laplace Transform of some functions

1. $L^{-1}\left\{\frac{1}{p}\right\} = 1$

2. $L^{-1}\left\{\frac{1}{p-a}\right\} = e^{at}$

3. $L^{-1}\left\{\frac{1}{p^n}\right\} = \frac{t^{n-1}}{(n-1)!}$ (if n is a positive integer) $\left[= t^{n-1}/\Gamma n \text{ otherwise}\right]$

4. $L^{-1}\left\{\frac{1}{(p-a)^n}\right\} = \frac{e^{at}\cdot t^{n-1}}{(n-1)!}$

5. $L^{-1}\left\{\frac{1}{p^2+a^2}\right\} = \frac{1}{a}\sin at$

6. $L^{-1}\left\{\frac{p}{p^2+a^2}\right\} = \cos at$

7. $L^{-1}\left\{\frac{1}{p^2-a^2}\right\} = \frac{1}{a}\sinh at$

7. $L^{-1}\left\{\frac{p}{p^2-a^2}\right\} = \cosh at$

Properties of Inverse Laplace Transform

1. **Linear Property:** If c_1 & c_2 are constants and $L\{F_1(t)\} = f_1(p)$, $L\{F_2(t)\} = f_2(p)$ then,

$$L^{-1}\{c_1 f_1(p) + c_2 f_2(p)\} = c_1 L^{-1}\{f_1(p)\} + c_2 L^{-1}\{f_2(p)\}.$$

2. **First Translation or Shifting Property:**

If $L^{-1}\{f(p)\} = F(t)$ then, $L^{-1}\{f(p-a)\} = e^{at}\cdot F(t)$

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3. Second Translation or Shifting Property: (2)

If $L^{-1}\{f(p)\} = F(t)$ then, $L^{-1}\{e^{-ap} f(p)\} = G(t)$ where

$$G(t) = \begin{cases} F(t-a) & t > a \\ 0 & t < a \end{cases}$$

4. Change of Scale Property

If $L^{-1}\{f(p)\} = F(t)$ then, $L^{-1}\{f(ap)\} = \frac{1}{a} F\left(\frac{t}{a}\right)$.

NOTE: Finding Inverse Laplace Transform sometimes becomes easier by breaking the expression using partial fractions.

Q. 1: Find ILT of (i) $\frac{2p+1}{p^2-4}$ (ii) $\frac{p+1}{p^2+p+1}$

Sol. (i) $L^{-1}\left\{\frac{2p+1}{p^2-4}\right\} = 2L^{-1}\left\{\frac{p}{p^2-4}\right\} + L^{-1}\left\{\frac{1}{p^2-4}\right\}$

$$= 2\cosh 2t + \frac{1}{2}\sinh 2t$$

(ii) $L^{-1}\left\{\frac{p+1}{p^2+p+1}\right\} = L^{-1}\left\{\frac{(p+\frac{1}{2})+\frac{1}{2}}{(p+\frac{1}{2})^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right\}$

$$= e^{-t/2} L^{-1}\left\{\frac{p+\frac{1}{2}}{p^2+(\sqrt{3}/2)^2}\right\} = e^{-t/2}\left[L^{-1}\left\{\frac{p}{p^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right\} + \frac{1}{2}L^{-1}\left\{\frac{1}{p^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right\}\right]$$

$$= e^{-t/2}\left[\cos\frac{\sqrt{3}}{2}t + \frac{1}{2}\times\frac{2}{\sqrt{3}}\sin\frac{\sqrt{3}}{2}t\right]$$

$$= \frac{1}{\sqrt{3}} e^{-t/2}\left[\sqrt{3}\cos\frac{\sqrt{3}}{2}t + \sin\frac{\sqrt{3}}{2}t\right].$$

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Q2: Find ILT of (i) $\frac{15}{p^2+4p+13}$ (ii) $\frac{e^{-p}}{\sqrt{p+1}}$

(iii) $\frac{14p+10}{49p^2+28p+13}$ (iv) $\frac{p-1}{p^2(p-7)}$

Sol. (i) $L^{-1}\left\{\frac{15}{p^2+4p+13}\right\} = L^{-1}\left\{\frac{15}{(p+2)^2+3^2}\right\} = e^{-2t}L^{-1}\left\{\frac{15}{p^2+3^2}\right\}$

$= e^{-2t}\cdot\frac{15}{3}\cdot\sin 3t = 5e^{-2t}\sin 3t$

(ii) $L^{-1}\left\{\frac{1}{\sqrt{p+1}}\right\} = e^{-t}L^{-1}\left\{\frac{1}{\sqrt{p}}\right\} = \frac{1}{\sqrt{\pi t}}e^{-t}$

$\Rightarrow L^{-1}\left\{\frac{e^{-p}}{\sqrt{p+1}}\right\} = \left.\begin{matrix}\frac{e^{-(t-1)}}{\sqrt{\pi(t-1)}} & t>1 \\ 0 & t<1\end{matrix}\right\} = \frac{e^{-(t-1)}}{\sqrt{\pi(t-1)}}u(t-1)$

(iii) $L^{-1}\left\{\frac{14p+10}{49p^2+28p+13}\right\} = L^{-1}\left\{\frac{14(p+2/7)+6}{49\left(p+\frac{2}{7}\right)^2+9}\right\}$

$= e^{-2t/7}L^{-1}\left\{\frac{14p+6}{49p^2+9}\right\} = e^{-\frac{2t}{7}}\left[\frac{14}{49}L^{-1}\left\{\frac{p}{p+\left(\frac{3}{7}\right)^2}\right\} + \frac{6}{49}L^{-1}\left\{\frac{1}{p+\left(\frac{3}{7}\right)^2}\right\}\right]$

$= e^{-\frac{2t}{7}}\left[\frac{2}{7}\cos\frac{3t}{7} + \frac{6}{49}\times\frac{7}{3}\sin\frac{3t}{7}\right]$

$= \frac{2}{7}e^{-2t/7}\left[\cos\frac{3t}{7} + \sin\frac{3t}{7}\right]$

(iv) $L^{-1}\left\{\frac{p-1}{p^2(p-7)}\right\} = L^{-1}\left\{\frac{-6}{49p} + \frac{1}{7p^2} + \frac{6}{49(p-7)}\right\}$

$= \frac{-6}{49}L^{-1}\left\{\frac{1}{p}\right\} + \frac{1}{7}L^{-1}\left\{\frac{1}{p^2}\right\} + \frac{6}{49}L^{-1}\left\{\frac{1}{p-7}\right\} = \frac{-6}{49} + \frac{t}{7} + \frac{6e^{7t}}{49}$

$\left[\frac{p-1}{p^2(p-7)} = \frac{A}{p} + \frac{B}{p^2} + \frac{C}{p-7} = \frac{Ap(p-7)+B(p-7)+Cp^2}{p^2(p-7)} \Rightarrow \begin{matrix} A+C=0 \\ B-7A=1 \\ -7B=-1\end{matrix}\right.$

$\left.\Rightarrow A = \frac{-6}{49},\ B = \frac{1}{7},\ C = \frac{6}{49}\right]$

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Inverse Laplace Transform of Derivatives

If $L^{-1}\{f(p)\} = F(t)$ then, $L^{-1}\{f^n(p)\} = (-1)^n t^n F(t)$

Multiplication by p

If $L^{-1}\{f(p)\} = F(t)$ and $F(0) = 0$ then, $L^{-1}\{p f(p)\} = F'(t)$

Division by p

If $L^{-1}\{f(p)\} = F(t)$ then, $L^{-1}\left\{\frac{f(p)}{p}\right\} = \int_0^t F(u)\,du$

Ex 1: Find ILT of (i) $\log\frac{(p+1)}{(p-1)}$ (ii) $\cot^{-1}\left(\frac{p+3}{2}\right)$

(iii) $\log\left[\frac{p^2+4p+5}{p^2+2p+5}\right]$ (iv) $\frac{1}{p^2+4}$

Sol. (i) $L^{-1}\left\{\log\left(\frac{p+1}{p-1}\right)\right\} = F(t)$ let

$\Rightarrow L^{-1}\left\{\frac{d}{dp}\left[\log(p+1) - \log(p-1)\right]\right\} = -tF(t)$

$\Rightarrow L^{-1}\left\{\frac{1}{p+1} - \frac{1}{p-1}\right\} = -tF(t)$

$\Rightarrow e^{-t} - e^{t} = -tF(t) \Rightarrow F(t) = \frac{e^t - e^{-t}}{2} = \frac{2\sinh t}{t}$

(ii) Let $L^{-1}\left\{\cot^{-1}\left(\frac{p+3}{2}\right)\right\} = F(t)$

$\Rightarrow L^{-1}\left\{\frac{d}{dp}\cot^{-1}\left(\frac{p+3}{2}\right)\right\} = -tF(t)$

$\Rightarrow L^{-1}\left\{\frac{-2}{(p+3)^2+4}\right\} = -tF(t)$

$\Rightarrow -e^{-3t}\sin 2t = -tF(t)$

$\Rightarrow F(t) = \frac{e^{-3t}\cdot\sin 2t}{t}$

(iii) Let $F(t) = L^{-1}\left\{ \log\left(\frac{p^2+4p+5}{p^2+2p+5}\right)\right\}$.

$$\Rightarrow -tF(t) = L^{-1}\left\{\frac{d}{dp}\left[\log(p^2+4p+5) - \log(p^2+2p+5)\right]\right\}$$

$$= L^{-1}\left\{\frac{2p+4}{p^2+4p+5} - \frac{2p+2}{p^2+2p+5}\right\}$$

$$= 2e^{-2t}L^{-1}\left\{\frac{p}{p^2+1}\right\} - 2e^{-t}L^{-1}\left\{\frac{p}{p^2+4}\right\}$$

$$= 2(e^{-2t}\cos t - e^{-t}\cos 2t)$$

$$\Rightarrow F(t) = \frac{2}{t}\left(e^{-t}\cos 2t - e^{-2t}\cos t\right).$$

(iv) $p^4+4 = (p^2+2)^2-(2p)^2 = (p^2-2p+2)(p^2+2p+2)$

$$\Rightarrow \frac{1}{p^4+4} = \frac{1}{(p^2-2p+2)(p^2+2p+2)} = \frac{1}{4p}\left[\frac{1}{p^2-2p+2} - \frac{1}{p^2+2p+2}\right]$$

Now, $L^{-1}\left\{\frac{1}{p^2-2p+2} - \frac{1}{p^2+2p+2}\right\} = L^{-1}\left\{\frac{1}{(p-1)^2+1^2} - \frac{1}{(p+1)^2+1^2}\right\}$

$$= e^{t}L^{-1}\left\{\frac{1}{p^2+1}\right\} - e^{-t}L^{-1}\left\{\frac{1}{p^2+1}\right\} = (e^t - e^{-t})\sin t$$

$$\Rightarrow L^{-1}\left\{\frac{1}{4p}\left(\frac{1}{p^2-2p+2} - \frac{1}{p^2+2p+2}\right)\right\} = \frac{1}{4}\int_0^t (e^t - e^{-t})\sin t\, dt$$

$$= \frac{1}{4}\left[\frac{e^t}{2}(\sin t - \cos t) - \frac{e^{-t}}{2}(-\sin t - \cos t)\right]$$

$$= \frac{1}{4}\left[\sin t\left(\frac{e^t+e^{-t}}{2}\right) - \cos t\left(\frac{e^t-e^{-t}}{2}\right)\right]$$

$$\Rightarrow L^{-1}\left\{\frac{1}{p^4+4}\right\} = \frac{1}{4}\left[\sin t\cosh t - \cos t \sinh t\right].$$

Q: 2 Find the inverse Laplace Transform of

(i) $\log\left(\frac{p+1}{p-1}\right)$ (ii) $\cot^{-1}\left(\frac{p+3}{2}\right)$ (iii) $L^{-1}\left\{\log\left(\frac{p^2+4p+5}{p^2+2p+5}\right)\right\}$

Sol. (i) let $L^{-1}\left\{\log\frac{(p+1)}{(p-1)}\right\} = F(t) \Rightarrow L^{-1}\left\{\frac{d}{dp}\left[\log(p+1)-\log(p-1)\right]\right\} = -tF(t)$

$\Rightarrow L^{-1}\left\{\frac{1}{p+1}-\frac{1}{p-1}\right\} = -tF(t) \Rightarrow e^{-t}-e^{t} = -tF(t) \Rightarrow F(t) = \frac{2\sinh t}{t}$

(ii) Let $L^{-1}\left\{\cot^{-1}\left(\frac{p+3}{2}\right)\right\} = F(t)$

$\Rightarrow L^{-1}\left\{\frac{d}{dp}\left\{\cot^{-1}\left(\frac{p+3}{2}\right)\right\}\right\} = -tF(t) \Rightarrow L^{-1}\left\{\frac{-2}{(p+3)^2+4}\right\} = -tF(t)$

$\Rightarrow -e^{-3t}\cdot\sin 2t = -tF(t) \Rightarrow F(t) = \frac{e^{-3t}\cdot\sin 2t}{t}$

(iii) let $L^{-1}\left\{\log\left(\frac{p^2+4p+5}{p^2+2p+5}\right)\right\} = F(t)$

$\Rightarrow L^{-1}\left\{\frac{d}{dp}\left[\log(p^2+4p+5)-\log(p^2+2p+5)\right]\right\} = -tF(t)$

$\Rightarrow L^{-1}\left\{\frac{2p+4}{p^2+4p+5}-\frac{2p+2}{p^2+2p+5}\right\} = -tF(t)$

$\Rightarrow L^{-1}\left\{\frac{2(p+2)}{(p+2)^2+1}-\frac{2(p+1)}{(p+1)^2+4}\right\} = -tF(t)$

$\Rightarrow 2e^{-2t}L^{-1}\left(\frac{p}{p^2+1}\right) - 2e^{-t}L^{-1}\left\{\frac{p}{p^2+4}\right\} = -tF(t)$

$\Rightarrow F(t) = \frac{2}{t}\left(e^{-t}\cos 2t - e^{-2t}\cdot\cos t\right)$

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Convolution Theorem:

If $L^{-1}\{f(p)\} = F(t)$ and $L^{-1}\{g(p)\} = G(t)$ then,

$$L^{-1}\{f(p)\cdot g(p)\} = F * G = \int_0^t F(u).\, G(t-u)\,du$$

Q:1 Use Convolution theorem to evaluate:

(i) $L^{-1}\left\{\frac{1}{p(p+1)(p+2)}\right\}$ (ii) $L^{-1}\left\{\frac{p^2}{(p^2+a^2)(p^2+b^2)}\right\}$

(iii) $L^{-1}\left\{\frac{16}{(p-2)(p+2)^2}\right\}$ (iv) $L^{-1}\left\{\frac{1}{p^3(p^2+1)}\right\}$

Sol (i) Let $f(p) = \frac{1}{p(p+1)}$ and $g(p) = \frac{1}{p+2}$

$\Rightarrow F(t) = L^{-1}\left\{\frac{1}{p(p+1)}\right\} = L^{-1}\left\{\frac{1}{p} - \frac{1}{p+1}\right\} = 1 - e^{-t}$

& $G(t) = L^{-1}\left\{\frac{1}{p+2}\right\} = e^{-2t}$

$\Rightarrow L^{-1}\left\{\frac{1}{p(p+1)(p+2)}\right\} = \int_0^t F(u).\, G(t-u)\,du = \int_0^t (1-e^{-u})\, e^{-2(t-u)}\,du$

$= e^{-2t}\int_0^t (e^{2u} - e^{u})\,du = e^{-2t}\left[\frac{e^{2u}}{2} - e^{u}\right]_0^t = e^{-2t}\left[\left(\frac{e^{2t}}{2} - e^{t}\right) - \left(\frac{1}{2} - 1\right)\right]$

$= \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}$

(ii) let $f(p) = \frac{p}{p^2+a^2}$, $g(p) = \frac{p}{p^2+b^2}$ $\Rightarrow F(t) = \cos at$, $G(t) = \cos bt$

$L^{-1}\left\{\frac{p^2}{(p^2+a^2)(p^2+b^2)}\right\} = \int_0^t \cos au.\, \cos b(t-u)\,du$

$= \frac{1}{2}\int_0^t \left[\cos\{(a-b)u + bt\} + \cos\{(a+b)u - bt\}\right] du$

$= \frac{1}{2}\left[\frac{a\sin at - b\sin bt}{a^2 - b^2}\right]$

(iii) $f(p) = \frac{1}{(p+2)^2}$, $g(p) = \frac{16}{p-2}$ $\Rightarrow$ $F(t) = t\,e^{-2t}$, $G(t) = 16e^{2t}$

$$\Rightarrow L^{-1}\{f(p).\,g(p)\} = \int_0^t F(u).\,G(t-u)\,du$$

$$= \int_0^t t\,e^{-2t}.16e^{2(t-u)}\,du = 16e^{2t}\int_0^t u\,e^{-4u}\,du = 16e^{2t}\left[\frac{u\,e^{-4u}}{-4} - \frac{e^{-4u}}{16}\right]_0^t$$

$$= 16e^{2t}\left[\frac{te^{-4t}}{-4} - \frac{e^{-4t}}{16} + \frac{1}{16}\right] = e^{2t} - e^{-2t}(4t+1).$$

(iv) let $f(p) = \frac{1}{p^2(p^2+1)} = \frac{1}{p^2} - \frac{1}{p^2+1}$ $\Rightarrow$ $F(t) = t - \sin t$

$g(p) = \frac{1}{p}$ $\Rightarrow$ $G(t) = 1$

$$\Rightarrow L^{-1}\left\{\frac{1}{p^3(p^2+1)}\right\} = \int_0^t F(u).\,G(t-u)\,du = \int_0^t (u - \sin u)\,1\,du$$

$$= \left[\frac{u^2}{2} + \cos u\right]_0^t = \frac{t^2}{2} + \cos t - 1$$

Q:2. Using Convolution theorem, prove that

$$L^{-1}\left\{\frac{p}{(p^2+a^2)^2}\right\} = \frac{t}{2a}\sin at$$

Sol. Let $f(p) = \frac{1}{p^2+a^2}$, $g(p) = \frac{p}{p^2+a^2}$ $\Rightarrow$ $F(t) = \frac{\sin at}{a}$, $G(t) = \cos at$

$$\Rightarrow L^{-1}\left\{\frac{p}{(p^2+a^2)^2}\right\} = \int_0^t \frac{\sin au}{a}\cos a(t-u)\,du =$$

$$= \frac{1}{2a}\int_0^t \left[\sin at + \sin(at - 2au)\right]du = \cancel{\frac{1}{a}\left[\frac{\cos at}{a} + \cos(at-\right.}$$

$$= \frac{1}{2a}\left[u\sin at + \frac{\cos(at-2au)}{-2a}\right]_0^t = \frac{1}{2a}\left[t\sin at\right] = \frac{t\sin at}{2a}$$

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Applications of Laplace Transform to Solve Differential Equations

Q:1. Using Laplace Transform, Solve the following differential equation:

(i) $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = e^{-t}.\sin t$ where $x(0)=0$ & $x'(0)=1$

(ii) $\frac{d^2y}{dt} + 9y = 6\cos 3t$; $y(0)=2$, $y'(0)=0$

(iii) $\frac{d^3y}{dt^3} - 3\frac{d^2y}{dt^2} + 3\frac{dy}{dt} - y = t^2e^t$; $y(0)=1$, $y'(0)=0$, $y''(0) = -2$

Sol (i) The given equation is: $x'' + 2x' + 5x = e^{-t}.\sin t$

Taking Laplace Transform on both sides,

$$L\{x'' + 2x' + 5x\} = L\{e^{-t}.\sin t\}$$

$$\Rightarrow \left[p^2\bar{x} - px(0) - x'(0)\right] + 2\left[p\bar{x} - x(0)\right] + 5\bar{x} = \frac{1}{(p+1)^2+1}$$

$(\bar{x} = L\{x\})$

$$\Rightarrow (p^2+2p+5)\bar{x} - 1 = \frac{1}{p^2+2p+2}$$ (Using given Conditions)

$$\Rightarrow \bar{x} = \frac{1}{(p^2+2p+2)(p^2+2p+5)} + \frac{1}{p^2+2p+2}$$

$$= \frac{1}{3}\left[\frac{1}{p^2+2p+2} + \frac{2}{p^2+2p+5}\right] = \frac{1}{3}\left[\frac{1}{(p+1)^2+1} + \frac{2}{(p+1)^2+2^2}\right]$$

Taking Inverse Laplace Transform on both sides,

$$x = \frac{1}{3}\left[e^{-t}\sin t + 2.\frac{1}{2}e^{-t}\sin 2t\right]$$

$$\Rightarrow x = \frac{e^{-t}}{3}\left[\sin t + \sin 2t\right]$$

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(ii) $y'' + 9y = 6\cos 3t$.

Taking Laplace transform on both sides,

$L\{y''\} + 9L\{y\} = 6L\{\cos 3t\}$.

$\Rightarrow [p^2\bar{y} - py(0) - y'(0)] + 9\bar{y} = \frac{6 \cdot p}{p^2+9}$

$\Rightarrow (p^2+9)\bar{y} - 2p = \frac{6p}{p^2+9} \Rightarrow \bar{y} = \frac{6p}{(p^2+9)^2} + \frac{2p}{p^2+9}$

Taking inverse Laplace Transform on both sides,

$y(t) = t\sin 3t + 2\cos 3t$.

(iii) $y''' - 3y'' + 3y' - y = t^2e^t$.

Taking Laplace Transform on both sides,

$L\{y'''\} - 3L\{y''\} + 3L\{y'\} - L\{y\} = L\{t^2 e^t\}$.

$\Rightarrow [p^3\bar{y} - p^2y(0) - py'(0) - y''(0)] - 3[p^2\bar{y} - py(0) - y'(0)]$ [where $\bar{y} = L\{y\}$]

$+ 3[p\bar{y} - y(0)] - \bar{y} = \frac{2}{(p-1)^3}$

$\Rightarrow (p^3\bar{y} - p^2 + 2) - 3(p^2\bar{y} - p) + 3(p\bar{y} - 1) - \bar{y} = \frac{2}{(p-1)^3}$

$\Rightarrow \bar{y} = \frac{(p-1)^2}{(p-1)^3} - \frac{p}{(p-1)^3} + \frac{2}{(p-1)^6}$

$= \frac{1}{(p-1)} - \frac{(p-1)+1}{(p-1)^3} + \frac{2}{(p-1)^6}$

$= \frac{1}{p-1} - \frac{1}{(p-1)^2} - \frac{1}{(p-1)^3} + \frac{2}{(p-1)^6}$

Taking Inverse Laplace Transform on both sides,

$y = e^t - te^t - \frac{t^2e^t}{2} + \frac{t^5e^t}{60} = \left(1 - t - \frac{t^2}{2} + \frac{t^5}{60}\right)e^t$

Q.2 Solve by Laplace Transform:

$$\frac{d^2y}{dt^2} + y = t\cos 2t \,, \quad t > 0 \text{ given that } y = \frac{dy}{dt} = 0 \text{ for } t = 0$$

Sol. Taking Laplace Transform on both sides,

$$L\{y''\} + L\{y\} = L\{t\cos 2t\}$$

$$\Rightarrow \left[p^2\bar{y} - p\,y(0) - y'(0)\right] + \bar{y} = -\frac{d}{dp}\left[\frac{p}{p^2+4}\right]$$

$$\Rightarrow (p^2+1)\,\bar{y} = \frac{p^2-4}{(p^2+4)^2}$$

$$\Rightarrow \bar{y} = \frac{p^2-4}{(p^2+1)(p^2+4)^2} = -\frac{5}{9}\cdot\frac{1}{p^2+1} + \frac{5}{9}\cdot\frac{1}{p^2+4} + \frac{8}{3(p^2+4)^2}$$

Taking Inverse Laplace Transform on both sides;

$$y = -\frac{5}{9}\sin t + \frac{5}{18}\sin 2t + \frac{8}{3}\,\frac{1}{16}\left(\sin 2t - t\cos 2t\right)$$

$$\Rightarrow y = -\frac{5}{9}\sin t + \frac{4}{9}\sin 2t - \frac{t}{3}\cos 2t$$